

Project 4: Preference Elicitation for Movie Recommendation

Human-Centric Artificial Intelligence — design report. We design and implement a user study comparing two interfaces for eliciting a new user's movie preferences: repeated pairwise choices (Design 1) and ranking lists of ten movies (Design 2). Both feed a linear utility model fit with a Plackett-Luce likelihood over the 4919 movies of the IMDB 5000 Movie Dataset. The interactive interface implementing Task 4 is linked from the project landing page.

Task 1 — Feature representation

The dataset contains only movie metadata, no ratings, so the feature vector x has to be built entirely from a movie's own attributes. We use five groups of features, chosen to cover the dimensions along which people typically describe liking or disliking a film:

- **Genre** (multi-hot over 26 genres) — the most direct signal of content type; a movie can carry several genre tags.
- **Content rating** (one-hot, bucketed to G / PG / PG-13 / R / NC-17 / Other) — tone and intended audience.
- **Production scale** — log-budget, log-gross, log-cast and log-director Facebook likes (star/production power).
- **Era and length** — release year and running time, z-scored.
- **Reach and reception** — log-vote-count (how mainstream the film is), IMDb score (critical consensus), and binary is-English / is-USA flags.

Concatenating these blocks gives a **42-dimensional** feature vector per movie. Skewed counts (votes, budget, gross, Facebook likes) are $\log_{10}$ -transformed before z-scoring so that a handful of blockbusters don't dominate the scale; missing numeric values are median-imputed and missing categorical values fall into an "Other"/"Not Rated" bucket rather than being dropped, so the full catalogue stays available for sampling. Exact duplicate (title, year) rows ($\approx 5\%$ of the raw file, mostly re-scraped re-releases) are removed so a participant is never asked to compare a movie against itself.

Deliberately excluded: an intercept/bias feature. The Bradley-Terry and Plackett-Luce likelihoods (Task 2) depend only on *differences* of utilities within a comparison set; a feature equal to 1 for every movie contributes an identical additive constant to every item's utility and cancels out of every comparison, so its weight cannot be estimated from choice data and including it would only waste one dimension.

Extraction is implemented in `project4/ml/data.py` (`load_movies` for cleaning, `build_features` for the vectors above) and cached once per process in `project4/services.py`.

Task 2 — From Bradley-Terry to rankings (Plackett-Luce)

The standard Bradley-Terry model gives the probability that movie i beats movie j from their utilities $U(x) = w^T x$:

$$P(i \blacksquare j) = \exp(U_i) / (\exp(U_i) + \exp(U_j))$$

Design 2 asks for a full ranking $i_1 \blacksquare i_2 \blacksquare \dots \blacksquare i_n$ of $n = 10$ movies rather than a single winner. We extend Bradley-Terry by treating the ranking as a sequence of “pick the best of what remains” choices: the top movie is chosen from all n candidates by a Bradley-Terry-style contest among all of them; the runner-up is then chosen the same way among the $n-1$ that remain; and so on down to the last pair. Multiplying the probabilities of these $n-1$ sequential choices gives the **Plackett-Luce model**:

$$P(i_1 \blacksquare \dots \blacksquare i_n) = \prod_{k=1}^{n-1} \exp(U_{i_k}) / \sum_{l=k}^n \exp(U_{i_l})$$

Setting $n = 2$ collapses this to exactly the Bradley-Terry formula above, so a pairwise comparison is simply a ranking of length 2. This means Design-1 and Design-2 responses can be combined into a single likelihood and fit jointly, which is what the live interface does when it computes a participant's final recommendations from all of their answers.

Fitting. We find w by maximizing this log-likelihood plus an L2 penalty $\lambda \|w\|^2$ (equivalently, a zero-mean Gaussian prior on w), using L-BFGS with an analytic gradient (`project4/ml/preference.py`). The penalty matters in practice: with only a few dozen comparisons and 34 features, unregularized maximum likelihood can diverge whenever the observed choices happen to be perfectly separable by some direction in feature space.

Assumption and limitation. Plackett-Luce inherits Luce's independence-of-irrelevant-alternatives property: the relative odds of i beating j never depend on which other items are present in the comparison set. This is a strong assumption — e.g. a participant's relative preference between two thrillers might shift once a comedy is added to the set as a “none of these either” option — but it is what makes pairwise and 10-way comparisons commensurable under one model, which the study needs in order to compare the two designs on equal footing.

Task 3 — User study design

Research question and hypotheses

RQ: For eliciting a new user's movie preferences under a fixed number of movies viewed, is a ranking-based interface or a pairwise-comparison interface more effective, and how do users experience each?

- **H1 (accuracy):** a preference vector fit from ranking responses predicts a participant's held-out choices at least as well as one fit from the same number of movie exposures worth of pairwise responses. Rationale: one ranking of 10 items logically implies up to $C(10,2) = 45$ pairwise orderings, so rankings should be more information-dense per movie shown.
- **H2 (subjective cost):** participants report ranking as more cognitively demanding than pairwise choice, even if it is more informative — i.e. we expect an accuracy/effort trade-off rather than one design dominating the other outright.

Design

Within-subjects, counterbalanced. Every participant completes both interfaces, in a randomly assigned order (pairwise $\rightarrow$ ranking, or ranking $\rightarrow$ pairwise), so each person acts as their own control for individual differences in taste and in general willingness to engage with the task. Counterbalancing order controls for practice and fatigue effects. The main threat specific to a within-subjects design is carry-over (the first task shapes how the participant approaches the second); we mitigate this by drawing an entirely fresh, non-overlapping set of movies for every phase, so there is nothing for the participant to consciously stay consistent with across phases.

Exposure-matched budget. The pairwise phase is 15 trials (30 movie views) and the ranking phase is 3 rankings of 10 movies (also 30 movie views), so both interfaces show the participant the same number of movies. This keeps the comparison about “information per movie shown” rather than “information per minute” or an arbitrarily chosen trial count for one condition.

Held-out evaluation block. After both elicitation phases, every participant completes 8 additional pairwise comparisons on freshly drawn movies, framed neutrally as “a few final comparisons” so they are answered in the same spirit as the rest of the study rather than as a test. These trials are never used to fit a preference vector — only to score one. All movies are drawn uniformly at random from the catalogue, as specified in the assignment brief; an adaptive/informative selection strategy is noted as an interesting extension but is out of scope here.

Independent and dependent variables

Variable	Type	Levels / operationalisation
Elicitation interface	Within-subject IV	Pairwise (Design 1) vs. Ranking (Design 2)
Presentation order	Between-subject IV	Pairwise-first vs. Ranking-first (counterbalanced)
Held-out predictive accuracy	DV (primary)	% of the 8 held-out pairwise choices correctly predicted by w fit on that phase's data alone
Held-out log-likelihood	DV (primary)	Plackett-Luce log-likelihood of held-out choices under each phase's w
Perceived difficulty	DV (subjective)	5-point Likert, one item per interface
Perceived confidence	DV (subjective)	5-point Likert: 'this reflected my real taste', per interface
Overall preference	DV (subjective)	Forced choice: pairwise / ranking / no preference
Completion time per phase	DV (behavioral)	Client-side timestamp, ms, recorded per trial

Materials

Each movie is shown as a card with title, year, up to 3 genres, up to 4 plot keywords, director, and top-3 billed cast — enough content information to form a genuine preference. **IMDb score and vote count are deliberately withheld** during both elicitation phases and only revealed on the final results screen: showing a visible “quality” number would anchor choices toward the more acclaimed/popular movie rather than the one the participant would personally rather watch, contaminating exactly the signal the study is trying to measure.

Procedure

- Landing page → consent screen (purpose, duration ≈8-10 min, voluntary participation, no personal data collected beyond an anonymous session id).
- Randomly assigned phase order; on-screen instructions before each phase.
- Phase A (15 pairwise trials, or 3 rankings of 10) → interstitial instructions → Phase B (the other interface).
- 8-trial held-out comparison block.
- Post-study questionnaire (difficulty, confidence, overall preference, free-text comments).
- Debrief screen: personalised top-5 recommendations computed from the participant's own responses (a concrete, motivating payoff for genuine engagement) and a completion code for crowdsourcing platforms.

Recruitment and sample size

Recruit via a platform such as Prolific, screening for fluent English and self-reported regular movie-watching (at least one movie per month), to avoid participants with too little basis for a preference. A rough power calculation for a paired comparison (e.g. Wilcoxon signed-rank on perceived difficulty, or a paired t-test on held-out accuracy) at a medium effect size ($d \approx 0.5$), $\alpha = .05$, power = .8 calls for roughly 34 participants; we would recruit **N = 45** to allow for exclusions. Compensation targeting a fair hourly rate (e.g. £1.50 for a ~9-minute study).

Quality control

One of the 15 pairwise trials is silently repeated later in the same phase with the identical pair; a participant who answers it inconsistently, combined with an implausibly fast median response time ($<1.5s$), is a candidate for exclusion before analysis. Session state and every individual response are persisted to the database as they happen, so partial/abandoned sessions remain available rather than being silently lost.

Ethics

No personal data is collected: participants are identified only by an anonymous, randomly generated session key. Participation is voluntary and participants may stop at any time with no way for us to trace an abandoned session back to them. A real deployment of this study would be submitted for institutional ethics review before recruiting; the consent screen implemented here reflects what that review would typically require (purpose, duration, voluntariness, data handling) but does not substitute for it.

Analysis plan

For each participant, fit two *independent* preference vectors — w_{pairwise} from only their pairwise-phase responses, w_{ranking} from only their ranking-phase responses — and score both against that participant's own held-out block (never used for fitting). Compare the two interfaces with a paired test (Wilcoxon signed-rank, participant as the unit of analysis) on: (1) held-out predictive accuracy, (2) held-out log-likelihood, (3) Likert difficulty/confidence ratings, and (4) time on task. Presentation order enters as a between-subjects check for carry-over (an order $\times$ interface interaction test); if no interaction is found, order is dropped and the two orders are pooled.

Interactive implementation of Task 4 (this protocol's actual participant-facing interface) is available from the Project 4 landing page.